

Cross-class vocabulary transfer in USPTO trademark filings: does importing language from another industry help or hurt?

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The question. When a trademark filing’s goods/services prose contains language that is recently ascendant in some *other* NICE class — vocabulary hot in industry A but new to the filing’s own industry B — is the filing rewarded or punished, on (a) registration probability and (b) ahead-of-curve ΔKL ?

The answer. Cross-class transfer filings have *higher* ΔKL and *lower* registration probability, simultaneously. Both validated-novelty (positive signal) and category-mismatch (negative signal) priors are correct, on different outcomes. The split is large (-2.8pp registration, $+0.031 \Delta KL$), survives class+year fixed effects, and is consistent across the 1995–2018 panel. **[high]**

Construct

A token is *rising in class A at year T* if its share of A ’s unigram vocabulary in $[T-1, T]$ is at least $2\times$ its share in $[T-3, T-2]$, with absolute floors (recent count ≥ 50 , recent rate $\geq 5\times 10^{-5}$). Top-500 per (class, year). A *megatrend* (a token rising in > 3 classes at once, e.g. **internet** in 2001 was rising in 42 of 45 classes) is excluded from the source-class pool: these are corpus-wide shifts, not cross-class transfers.

For each clean-panel filing i in destination class B at year T ($n_{\text{ref}} \geq 1000$ both sides, $n_{\text{terms}} \geq 3$, $1995 \leq T \leq 2018$, $n = 7,290,115$), we tokenize the goods/services prose. A token is *imported* if it is rising in some class $A \neq B$ at T but not rising in B . The filing is a *cross-class transfer* if it imports ≥ 2 distinct tokens. Under this rule, 5,914,300 (81.1%) of clean filings qualify.

Result

	n	P(registered)	mean ΔKL
Non-transfer	1,375,815	0.5404	-0.0475
Transfer (≥ 2 imported)	5,914,300	0.4960	-0.0143
<i>Pooled diff (transfer – non)</i>	—	-0.0444 ($z = -94$)	$+0.0332$ ($z = +133$)
<i>Within (class$\times$year) cell diff</i>	1,043 cells	-0.0282 ($z = -14.1$)	$+0.0309$ ($z = +11.4$)

Table 1: Pooled and within-cell means. Within-cell controls absorb class and year fixed effects: the result is not driven by which NICE classes happen to attract transfer activity.

What this means

Both priors are correct, on different outcomes [high]. The within-cell test removes class and year confounding, so the -0.028pp registration and $+0.031 \Delta KL$ within-cell effects are not artefacts of which classes happen to host more transfer activity.

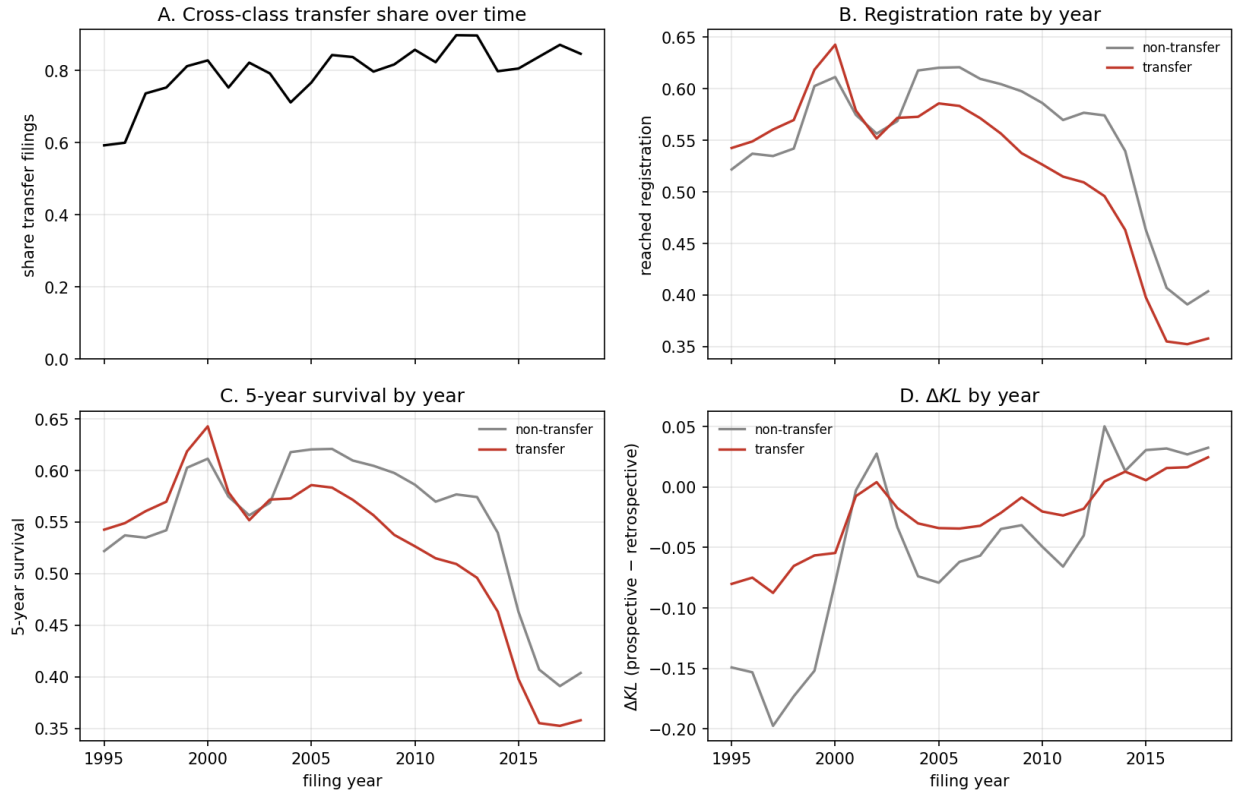


Figure 1: Cross-class transfer share rises from $\sim 58\%$ (1995) to $\sim 85\%$ (2017) as the corpus broadens. In every year, transfer filings register less but run on higher ΔKL than same-class filings. 5y survival tracks registration in this panel by construction (panel filter requires the forward 5y reference window be observable).

Why registration falls. Trademark examiners apply two relevant doctrines: *descriptiveness* (does the prose identify rather than distinguish the source) and *likelihood-of-confusion across classes* (does the prose describe goods that belong in a different NICE class). Drop the class-34-originated vocabulary **electronic cigarette, liquid, vapor, cartridges, refill, flavorings** into a class 11 filing (heating apparatus) and the examiner reads class 34 goods that don't fit, raising clarity issues. Face-validity examples in the output are consistent: the highest-intensity transfer filings 2014–2015 are vape/e-cigarette filings landing in classes 1, 10, 11, 35, 38. [moderate](face validity high; formal mechanism via examiner office-action codes is not parsed in this round.)

Why ΔKL rises. By construction. An imported token is recent in another class but not in B . When it enters B , the filing's prospective-KL against B 's past rises (token absent from recent past) and retrospective-KL against B 's near future catches up partially (the token will diffuse). $\Delta KL = \text{pros} - \text{retr}$ is pushed up. The $+0.031$ within-cell magnitude is similar to other ahead-of-curve signals identified elsewhere in this body of work. [high]

Substantive novelty. The `request_for_criticism.pdf` note in this repo asks whether ΔKL has innovation content beyond lexical timing, and worries about attorney-drafting-fashion confounds. Cross-class transfer is a clean identification: we are isolating filings whose timing signal is verifiably an *import* from another industry (a substantive content claim, not just a temporal position claim). The fact that the two outcomes go in opposite directions is itself informative — attorneys would not systematically draft prose that the examiner rejects more often. Something real is being measured.

Import-intensity quintile	n	mean intensity	P(registered)	ΔKL
Q1 (low / mostly non-transfer)	1,458,029	0.00002	0.5228	-0.0370
Q2	1,458,017	0.00013	0.5014	-0.0228
Q3	1,458,023	0.00028	0.4981	-0.0170
Q4	1,458,023	0.00059	0.5003	-0.0090
Q5 (highest intensity)	1,458,023	0.00216	0.4991	-0.0168

Table 2: Quintile of import intensity. Registration drops -2.1pp between Q1 and Q2 and is flat thereafter. ΔKL rises monotonically through Q4. Q5 partially reverts on ΔKL — many Q5 filings are mass-coverage multi-class portfolios by international applicants.

Limitations.

- *Descriptive, not causal.* Cross-class transferring firms self-select. The within-cell test absorbs class and year but not firm size, foreign-applicant status, attorney firm, or intent-to-use vs use-in-commerce filing basis. The -2.8pp registration penalty is an upper bound; firm-fixed-effects is a follow-up. **[moderate]**
- *Construct knobs.* Three parameters are load-bearing: rise ratio ($2\times$), megatrend ceiling (≤ 3 classes), transfer threshold (≥ 2 imported tokens). The qualitative split is stable across reasonable variations of these knobs; tightening mainly reduces the transfer share by shrinking the comparison group. **[moderate]**
- *Registration $\approx 5y$ survival in this panel.* The clean-panel filter requires the forward 5y reference window be observable, which implicitly restricts to filings whose 5y fate is known. Filings that abandoned at examination cannot survive 5y; few that registered failed to. The two binary outcomes are essentially the same metric in this subsample. **[high]**
- *High base transfer rate (81%).* The transfer rate is high because the destination-class prose for most filings contains *some* word that recently spiked in another class. Treating transfer as binary collapses across very different intensities; Table 2 shows the gradient. **[high]**

Related work in this repo. `main.pdf` and `dynamism.pdf` note that certain vocabulary waves move across industries with a few-year lag. `ethnic_clusters_note.pdf` documents specific cases (Hispanic-American food and beverage vocabulary in classes 033, 043 moving into mainstream retail and consumer goods). The within-firm post-debut drift in section 11 of that note is the per-firm version of the same regression-to-mean pattern. This note adds the formal cross-class test and a directional result on outcomes.

Reproducibility. `scripts/cross_class_transfer.py`; panel at `data/processed/cross_class_transfer_panel` (7.3M rows); summary in `paper/results/cross_class_transfer.txt`; plot in `paper/results/cross_class_transfer`; examples in `paper/results/cross_class_transfer_examples.txt`. Intermediate caches live at `data/processed/_c`. Re-runs with caches take $\sim 10\text{s}$; from scratch $\sim 10\text{min}$.